Minimum Cost to Connect Sticks(medium)

Problem Statement

Given a collection of sticks with different lengths. To combine any two sticks, there's a cost involved, which is equal to the sum of their lengths.

Connect all the sticks into a single one with the minimum possible cost. Remember, once two sticks are combined, they form a single stick whose length is the sum of the lengths of the two original sticks.

Examples

1.
   - **Input:** [2, 4, 3]
   - **Expected Output:** 14
   - **Justification:** Combine sticks 2 and 3 for a cost of 5. Now, we have sticks [4,5]. Combine these at a cost of 9. Total cost = 5 + 9 = 14.

2.
   - **Input:** [1, 8, 2, 5]
   - **Expected Output:** 30
   - **Justification:** Combine sticks 1 and 2 for a cost of 3. Now, we have sticks [3, 8, 5]. Combine 3 and 5 for a cost of 8. Now, we have sticks [8,8]. Combine these for a cost of 16. Total cost = 3 + 8 + 16 = 27.

3.
   - **Input:** [5, 5, 5, 5]
   - **Expected Output:** 40
   - **Justification:** Combine two 5s for a cost of 10. Do this again for another cost of 10. Now, we have two sticks of 10 each. Combine these for a cost of 20. Total cost = 10 + 10 + 20 = 40.

Try it yourself

Try solving this question here:

Java

```java
import java.util.PriorityQueue;

public class Solution {
    public int connectSticks(int[] sticks) {
        int cost = 0;
        // ToDo: Write Your Code Here.
        return cost;
    }
}
```

Test Case | Run Code Result | Stdout

[2, 4]

Use Example Testcases | Run | Submit

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Solution: Minimum Cost To Connect Sticks(Medium)

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